

POWER APPS

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Agenda

- What is Power Platform?
- Fusion development
- Power Apps introduction
 - What is Power Apps?
 - Why use Power Apps?
 - What can you build
 - Power Apps components
 - Canvas apps
 - Dataverse
 - Model-driven apps
 - Integration with Power Automate
- Hackathon labs: Power App in a day

What is Power Platform?

- Collection of low/no-code tools for rapidly building custom business apps
 - Turns business needs into secure, connected applications
- Main components
 - Power Apps
 - Builder for creating custom mobile and web business applications
 - Power Automate
 - Workflow service that automates repetitive tasks and orchestrates activities across services using connectors
 - Power BI
 - Provides interactive data visualization Business Intelligence tools
 - Power Pages
 - Platform for designing, hosting and managing secure business websites
 - Copilot Studio
 - Tool for building AI-powered conversational agents

Fusion development

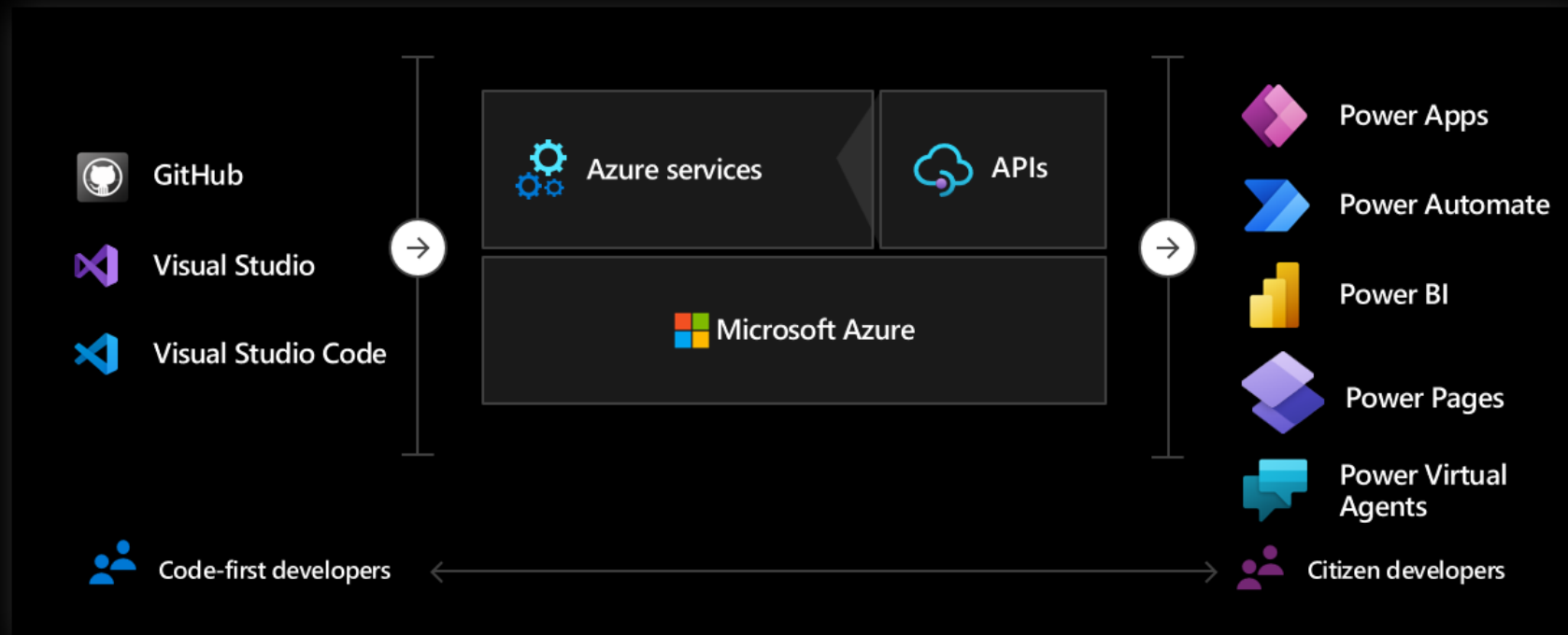
- “Distributed and multidisciplinary digital business teams that blend technology and other types of domain expertise.”
 - 2019 Gartner Digital Business Teams Survey
- Enables building apps
 - Better
 - Faster
- Goal: Scale application development through low-code
 - Empower people with different skill sets

Fusion team roles

- Citizen (low-code) developers
 - Non-technical
 - Knowledgeable about ways to improve business processes
 - Use visual drag/drop tools
- Professional developers
 - Highly technical
 - Write code for custom connectors, integrating w/external data, applying business logic
 - Use tools like Visual Studio/VS Code
- IT professionals/DevOps engineers
 - Responsible for application lifecycle management (ALM) of business apps

Fusion team development

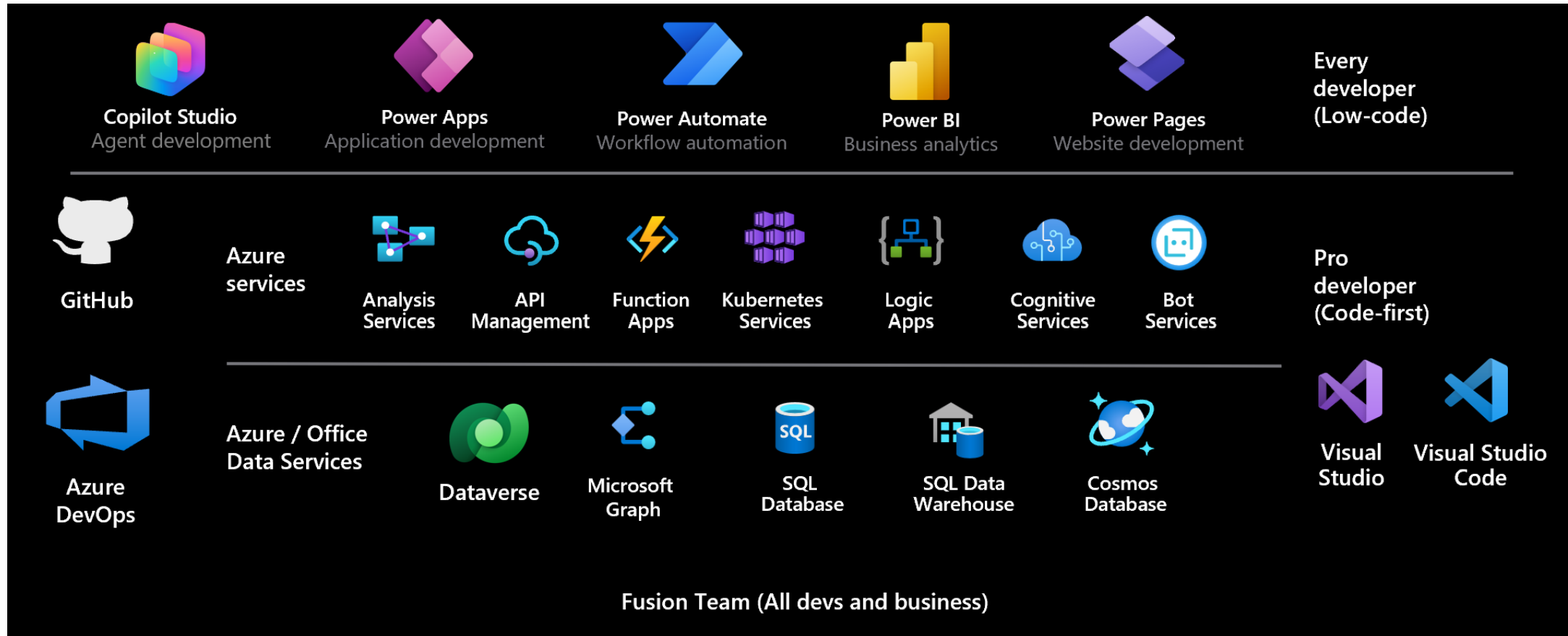
Fusion Team Development



Citizen developer tools

- Designed to make the citizen developer more productive
- Created
 - By professional developer/IT professional
 - Within a secure and governed environment
- Should be usable by everyone for low-code tasks
 - Not limited to citizen developers
 - E.g., Visual tooling for creating apps

Fusion team development tools



What is Power Apps?

- Low-code platform for providing the business app user experience
 - Mobile & web
- Software suite consisting of
 - Apps
 - Services
 - Data platform
 - Connectors

Why Power Apps?

- Build quickly
 - Use visual designers, formulas, and reusable components
- Connect data using
 - Dataverse
 - Many cloud or on-prem sources
- Run anywhere
 - Web browser
 - Mobile devices
 - Phone (iOS and Android)
 - Tablets

What can you build?

Look for manual, repetitive, data-driven processes that need a better experience.

Replace manual work

Forms, spreadsheets, paper

Track activities

Assets, requests, issues

Modernize legacy apps

Departmental tools

Capture field data

Inspections, visits, inventory

Standardize approvals

Rules, routing, status

Extend existing systems

Focused task experiences

Candidate test: Is the process manual, repetitive, data-driven, or fragmented across tools?



Power Apps components

- Canvas apps
 - Start from a blank canvas, build interface & connect to data sources
- Microsoft Dataverse
 - Data platform for Power Apps that stores and models business data
- Model-driven apps
 - Start with data model & build up to model forms, views & other components
- Plans
 - Copilot-first tool for building end-to-end business solution
- Copilot
 - Create an app & data model using Copilot

Canvas apps

- Start from blank design surface for building user experience
 - Drag and drop components to build UI
- Connect
 - > 200 data sources
 - Works on all major browsers and mobile devices
- Share
 - Securely within organization
- Embed
 - Directly in SharePoint, Power BI and Teams

Dataverse

- Store business data
 - Secure cloud-based storage for Power Platform applications
 - Standard and custom tables for business information
- Model relationships
 - Connect tables through relationships
 - Create a shared source of truth across applications
- Manage business logic
 - Business rules help enforce data consistency
 - Calculated and derived values reduce application complexity
- Secure
 - Role-based security controls access to data
 - Supports organizational security boundaries
- Integrate
 - Used by Power Apps, Power Automate, and Dynamics 365
 - Accessible through connectors and APIs
- Scale
 - Designed for enterprise applications and large data volumes
 - Centralized platform for data, automation, reporting, and AI solutions

Model-driven apps

- Start from the data model
 - Build on Dataverse tables, relationships, forms, and views
- Generate the application
 - Navigation, forms, and views are generated from app components
 - Focus on data and business processes rather than pixel-perfect layout
- Manage processes
 - Business process flows guide users through work
 - Business rules enforce data consistency
- Secure
 - Role-based security controls access to data and application features
- Share
 - Securely within the organization
- Extend
 - Add charts, dashboards, Power Automate flows, and custom pages

Power Automate

- Automate processes
 - Trigger workflows from events, schedules, or user actions
 - Reduce manual and repetitive tasks
- Connect systems
 - Integrate Microsoft 365, Power Platform, and external services
 - Hundreds of prebuilt connectors available
- Manage approvals
 - Route requests to approvers automatically
 - Track decisions and outcomes
- Orchestrate workflows
 - Combine data, business rules, and actions
 - Coordinate activities across multiple systems
- Respond to events
 - Start flows when records are created or updated
 - Send notifications and update related systems
- Extend solutions
 - Add automation to Power Apps and Dataverse
 - Support end-to-end business processes

Canvas vs. Model-Driven apps

Which app type should you use?

Choose based on the experience the process needs, not which designer looks easier.

START WITH The user experience

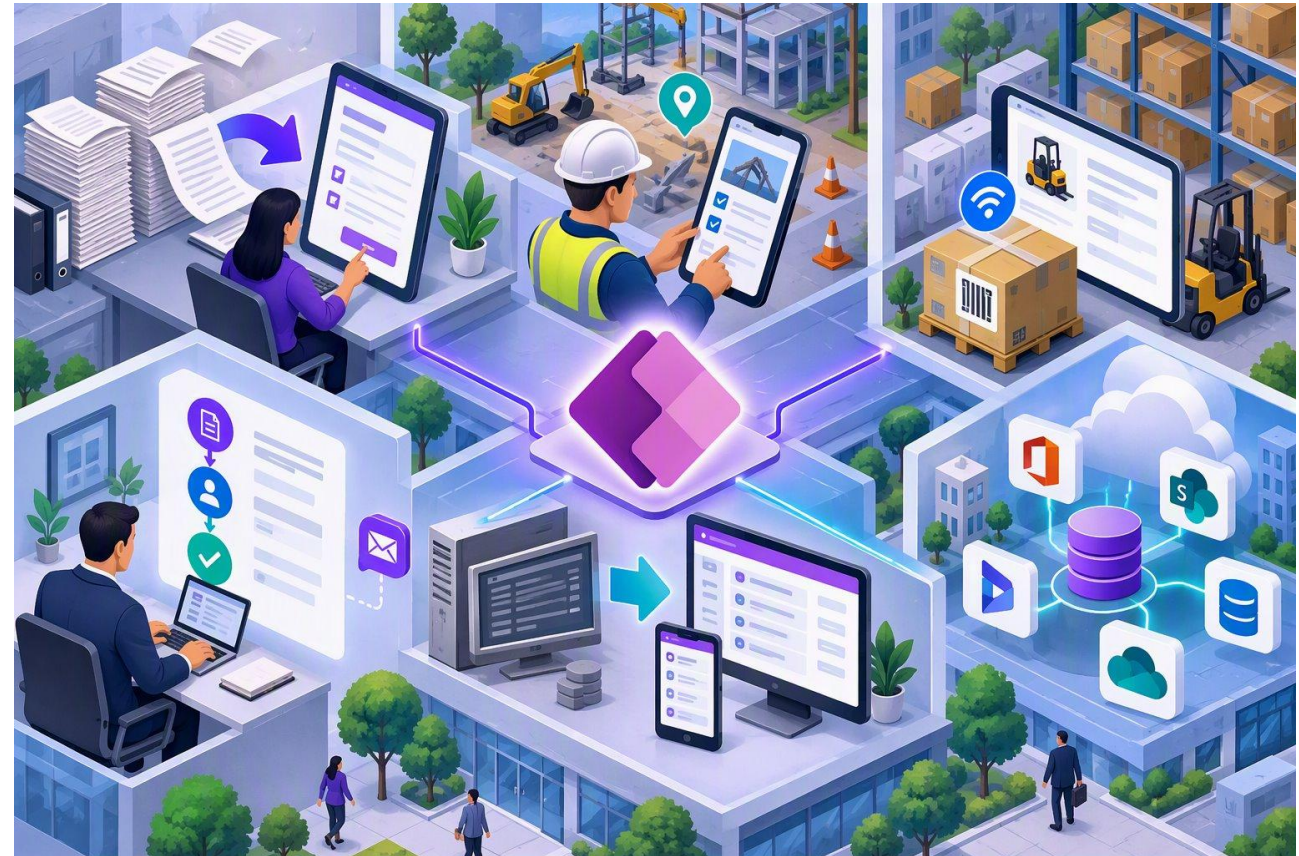
BEST WHEN Layout and interaction matter

TYPICAL FIT Mobile and task-focused apps

START WITH The data model

BEST WHEN Processes are data-dense

TYPICAL FIT Case and record management



They can also work together in one solution.

Power Platform components summary

- Power Apps
 - Canvas app provides the employee experience
 - Model-driven app provides the operations experience
- Dataverse
 - Stores and models the business data
 - Other connected data systems can also be used
- Power Automate
 - Handles approvals, notifications, and workflow steps

Lab

What will we build today?

One business solution, built progressively from data to app to automation.

- Experience** Build a canvas app for everyday users
- Model** Create Dataverse tables and relationships
- Process** Create a model-driven app for operations
- Automate** Add business logic and an approval flow
- Deliver** Test, publish, and share the solution



Lab details

- Create a machine ordering app with Power Apps
 - Online workshop from App in a Day series of courses
 - <https://learn.microsoft.com/en-us/training/paths/power-apps-online-workshop/>
- Exercises
 - Create a machine ordering Power Apps canvas app
 - Estimated time: 1 hr 24 min
 - Create a Dataverse table for the machine ordering app
 - Estimated time: 1 hr 33 min
 - Create a model-driven Power Apps app for machine ordering
 - Estimated time: 1 hr 20 min
 - Integrate a Power Apps app with Power Automate
 - Estimated time: 1 hr 17 min

Lab domain – Coffee machine ordering

- Employees request new coffee machine for break room every few years
- Organization constrained by lack of development resources
 - No traditional developers available to create custom solution
- Goal: build a customized app for machine ordering
 - Runs on the web and mobile devices
 - Streamline machine order and approval
- Solution: Use Power Platform to empower tech savvy business users to create app
 - Power Apps
 - Application UI and interaction logic
 - Dataverse
 - Stores coffee machine order data
 - Power Automate
 - Stores machine order data